

Mars Rover Pathfinder

Autonomous Navigation using a Markov Decision Process (MDP)

Technical Report

Prepared for academic purposes

1. Introduction

Autonomous space exploration is a key domain of intelligent robotics, particularly for planetary missions such as Mars exploration. Martian rovers must navigate unknown environments that can be hazardous and subject to uncertainty, while still reaching mission goals efficiently.

In this project, we study the autonomous navigation problem for a Mars rover, modeled as a Markov Decision Process (MDP). This framework formally represents sequential decision-making in a stochastic environment, where actions do not always lead to the intended outcome.

The main objective is to show how an autonomous agent can compute an optimal navigation policy that maximizes cumulative reward, using the Value Iteration algorithm.

2. Problem Statement and Objectives

2.1 Problem Statement

The problem can be formulated as follows: How can a Martian rover choose, at each time step, the optimal action to reach a target destination, while accounting for motion uncertainty and environmental constraints?

The main challenges are:

- Stochastic transitions (slippage, motion errors).
- The need to plan over multiple steps.
- Optimizing a global objective rather than an immediate gain.

2.2 Project Objectives

The objectives of the project are:

- Model a realistic autonomous navigation problem using an MDP.
- Implement the Value Iteration algorithm.
- Compute an optimal policy.
- Analyze rover behavior through simulation.
- Identify limitations of the model and propose improvements.

3. Markov Decision Process Modeling

The navigation problem is modeled as an MDP defined by the tuple (S, A, P, R, γ) .

3.1 States (S)

States represent the rover's possible positions in a 2D grid world. Each cell of the grid corresponds to a distinct state. This abstraction simplifies the spatial problem, facilitates decision analysis, and focuses on decision-making rather than physical complexity.

3.2 Actions (A)

From each state, the rover can take a discrete set of actions: move up, down, left, or right. This is the classical action space for grid-based navigation problems.

3.3 Stochastic Transitions (P)

Unlike deterministic motion, transitions are stochastic: an action succeeds with a given probability, or the rover may end up in an unintended neighboring state (slippage, imprecision). This reflects Martian terrain uncertainty and actuator limitations.

3.4 Reward Function (R)

The reward function is designed to penalize each movement (energy cost) and encourage reaching the final goal. This structure pushes the rover to avoid unnecessary steps and to seek short, efficient paths.

3.5 Discount Factor (gamma)

The discount factor gamma ($0 < \gamma < 1$) prioritizes near-term future rewards, ensures algorithm convergence, and models increasing uncertainty over long horizons.

4. Solution Method: Value Iteration

4.1 General Principle

Value Iteration is based on the optimal Bellman equation:

$$V(s) = \max_a \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V(s')]$$

It computes the optimal value of each state by propagating rewards through the state space.

4.2 Convergence Process

The algorithm proceeds through successive iterations:

- Initialize state values.
- Update values using the Bellman optimality backup.
- Stop when the variation in values becomes negligible.

Because the state space is finite and $\gamma < 1$, convergence is guaranteed.

4.3 Extracting the Optimal Policy

Once values stabilize, the optimal policy is obtained by selecting, for each state, the action that maximizes the expected return.

5. Simulation and Results

Simulation of the rover following the computed optimal policy shows that:

- The rover converges toward the goal.

- Trajectories avoid penalized zones.
- The policy remains stable despite stochastic transitions.

States near the goal have higher values, explained by the backward propagation of the terminal reward.

6. Critical Analysis of the Model

6.1 Strengths

- Rigorous modeling of a realistic problem.
- Correct use of fundamental MDP concepts.
- Readable and functional implementation of Value Iteration.
- A simulation that supports interpretation and validation.

6.2 Identified Limitations

Despite its qualities, the model has limitations:

- Static and simplified environment.
- Perfect observability.
- No detailed energy constraints.
- Small state space.

These simplifications are intentional, in order to focus on understanding the decision-making mechanism.

7. Perspectives and Future Work

Several extensions can be considered:

- Integrate dynamic obstacles.
- Introduce explicit energy management.
- Move to POMDPs for imperfect sensors.
- Compare with other methods (Policy Iteration, Q-learning).
- Use Deep Reinforcement Learning techniques.

8. Conclusion

This project demonstrates the effectiveness of Markov Decision Processes for solving autonomous navigation problems in uncertain environments. Although the framework is intentionally simplified, it highlights the fundamental mechanisms of optimal sequential decision-making.

It provides a solid foundation for more complex systems and naturally leads toward advanced AI approaches for autonomous robotics.